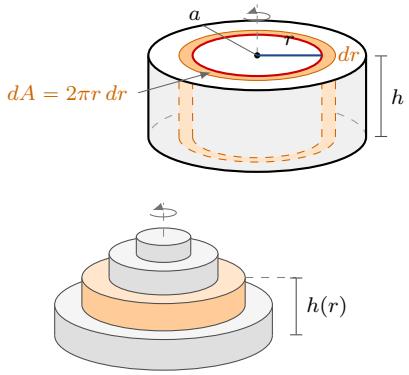
L4.3a Cylindrical Shells

Volume along r — what you need to know

VOLUME OF A CYLINDER, RING BY RING



Cut the base of a cylinder of radius a and height h into thin rings. The ring at radius r has width dr and area $dA = 2\pi r\,dr$, and the cylinder stands a thin shell of height h on it. Add the shells from the center out:

$$V = h\int_0^a dA = h\int_0^a 2\pi r\,dr = 2\pi h\left[\tfrac{r^2}{2}\right]_0^a = \pi a^2 h \checkmark$$

If the height changes with the radius, each ring carries its own height $h(r)$:

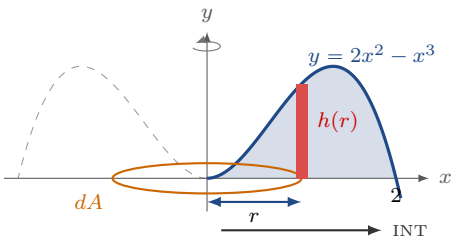
$$\boxed{V = \int_{\text{inner}}^{\text{outer}} h(r)\,dA, \qquad dA = 2\pi r\,dr}$$

r is the distance from the axis of rotation. The limits are the inner and outer radius of the solid.

"$h(r)$ is the weighting function on dA: how thick the solid is, along the axis, over the ring at radius r"

VOLUME BY CYLINDRICAL SHELLS

Ex. Rotate the region under $y = 2x^2 - x^3$, $0 \leqslant x \leqslant 2$, about the y-axis.



① **Draw a representative shell,** a thin upright rectangle in the region, parallel to the spin axis. Spun about the axis, it stands on the ring $dA = 2\pi r\,dr$.

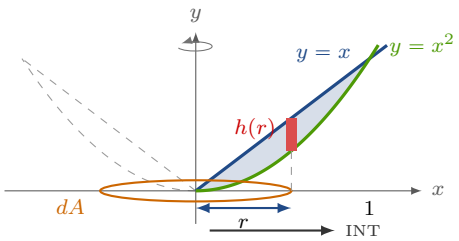
② r **is the distance** from the spin axis to the rectangle: $r = x$.

③ $h(r)$ **is the length** of the rectangle inside the shaded region: from $y = 0$ up to the curve, $h(r) = 2r^2 - r^3$.

④ **Limits:** the rectangle sweeps from the axis, inner 0, to the edge, outer 2.

$$V = \int_0^2 \left(2r^2 - r^3\right) 2\pi r\,dr = 2\pi\int_0^2 2r^3 - r^4\,dr = 2\pi\left[\tfrac{1}{2}r^4 - \tfrac{1}{5}r^5\right]_0^2 = 2\pi\left(8 - \tfrac{32}{5}\right) = \boxed{\tfrac{16\pi}{5}}$$

SHELLS FOR A REGION BETWEEN TWO CURVES



Ex. Rotate the region between $y = x$ and $y = x^2$ about the y-axis.

① **Draw a representative shell,** an upright rectangle from the lower curve to the upper.

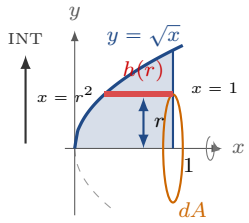
② r **is the distance** from the spin axis: $r = x$.

③ $h(r)$ **is top minus bottom.** Test $x = \frac{1}{2}$: $\frac{1}{2} > \frac{1}{4}$, so $y = x$ is on top: $h(r) = r - r^2$.

④ **Limits:** the curves cross at $x = 0$ and $x = 1$: inner 0, outer 1.

$$V = \int_0^1 (r - r^2)\, 2\pi r\, dr = 2\pi \int_0^1 r^2 - r^3\, dr = 2\pi \left[\frac{r^3}{3} - \frac{r^4}{4}\right]_0^1 = 2\pi \left(\frac{4-3}{12}\right) = \boxed{\frac{\pi}{6}}$$

SHELLS ABOUT THE x-AXIS



Ex. Rotate the region under $y = \sqrt{x}$, $0 \leqslant x \leqslant 1$, about the x-axis.

① **Draw a representative shell,** a sideways rectangle. Spun, it is a tube on its side, standing on the ring $dA = 2\pi r\, dr$.

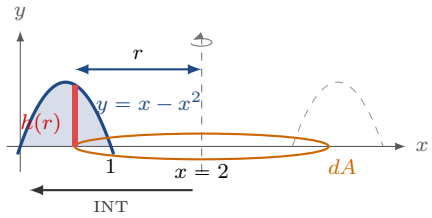
② r **is the distance** from the x-axis: $r = y$.

③ $h(r)$ **is right minus left.** It enters at the curve, $r = \sqrt{x}$ so $x = r^2$, and leaves at $x = 1$: $h(r) = 1 - r^2$.

④ **Limits:** inner 0, outer 1.

$$V = \int_0^1 (1 - r^2)\, 2\pi r\, dr = 2\pi \int_0^1 r - r^3\, dr = 2\pi \left[\frac{r^2}{2} - \frac{r^4}{4}\right]_0^1 = \boxed{\frac{\pi}{2}}$$

SHELLS ABOUT A LINE THAT IS NOT AN AXIS



Ex. Rotate the region between $y = x - x^2$ and $y = 0$ about the line $x = 2$.

① **Draw a representative shell,** an upright rectangle.

② r **is the distance** from $x = 2$. The region is left of the line: $r = 2 - x$, so $x = 2 - r$.

③ $h(r)$ **is top minus bottom,** written in r: $h(r) = x - x^2 = (2 - r) - (2 - r)^2 = -r^2 + 3r - 2$.

④ **Limits:** $x = 1$ is nearest the line, $r = 1$; $x = 0$ is farthest, $r = 2$.

$$V = \int_1^2 (-r^2 + 3r - 2)\, 2\pi r\, dr = 2\pi \int_1^2 -r^3 + 3r^2 - 2r\, dr = 2\pi \left[-\frac{r^4}{4} + r^3 - r^2\right]_1^2 = \boxed{\frac{\pi}{2}}$$

WATCH OUT

✗ ~~$h(r)$ is the function $f(x)$~~ → $h(r)$ **is top minus bottom, or right minus left**

✗ ~~about the x-axis, $r = x$~~ → r **is the distance from the axis:** $r = y$

✗ ~~about $x = 2$, $r = x - 2$~~ → **the region is left of the axis:** $r = 2 - x$